1.

Use Wireshark to capture network traffic while you load www.spotify.com in your browser, then identify and note down at least two DNS packets and two HTTP or HTTPS packets in the capture.

ANS:-

Wireshark Traffic Analysis:

DNS Packets:

Packet	Protocol	Purpose
1	DNS	Resolves www.spotify.com to an IP address.
2	DNS	Resolves another Spotify-related domain to an IP address.

HTTP/HTTPS Packets:

Packet	Protocol	Purpose
1	HTTPS	Secure communication between the browser and Spotify.
2	HTTPS	Transfers encrypted web data between the browser and Spotify.

Result: Two DNS packets and two HTTPS packets were identified while loading www.spotify.com in Wireshark.

2.

Simulate a DNS lookup for www.flipkart.com using the nslookup command in your terminal and write down the IP address returned along with a brief explanation of what this result means.

ANS:-

Simulate a DNS lookup for www.flipkart.com using the nslookup command in your terminal and write down the IP address returned along with a brief explanation of what this result means.

DNS Lookup Using NSLookup:

Command Used: nslookup www.flipkart.com

Result:

- IP Address: 163.53.76.86

Explanation: ☑ The returned IP address is the server address associated with www.flipkart.com. It shows that DNS successfully converted the website name into an IP address that the computer can use to connect to the website.

3.

Explain the difference between HTTP and HTTPS by listing 3 key differences, and give a real-world example of an app or website you use that relies on HTTPS for secure communication.

ANS:-

HTTP vs HTTPS:

Point	HTTP	HTTPS
Security	Less secure; data is not encrypted.	More secure; data is encrypted.
Protection	Easier for attackers to read or modify data.	Protects data from being easily intercepted or changed.
Usage	Used for basic, non-sensitive web communication.	Used for secure activities like logins and payments.

Real World Example: Instagram uses HTTPS to securely protect user login details and other data during communication.

4.

Open Wireshark and filter for DNS protocol traffic. Try searching for a new website (one you haven't visited recently) in your browser and describe the sequence of DNS queries and responses you observe. Hint: Look for packets with 'Standard query' and 'Standard query response' in the Info column.

ANS:-

DNS Query and Response in Wireshark:

DNS Sequence Observed

1. Standard query: The browser sends a DNS query to find the IP address of the new website.
2. Standard query response: The DNS server replies with the website's IP address.
3. The browser uses this IP address to connect to the website.

Result: The DNS packets showed a clear query → response sequence, confirming that DNS resolved the website name into an IP address.

5.

Pick any one protocol from DNS, DHCP, FTP, or SNMP (other than HTTP/HTTPS) and write a short scenario describing how it would be used in a real-world app or service you use (for example, how DHCP helps your phone connect to WiFi when you open Instagram at a cafe).

ANS:-

Real-World Use of DHCP:

- **Protocol Chosen:** DHCP (Dynamic Host Configuration Protocol)

Scenario: When I connect my phone to Wi-Fi at a café and open Instagram, DHCP automatically assigns my phone an IP address, subnet mask, and default gateway. This allows my phone to communicate with the internet without manually entering network settings.